Additive Geometry I

Mathematics Seminar at Kyiv School of Economics

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1 Lattices

In these notes, we follow [TV06].

Definition 1.1. *We define the following notions:*

(i) *A **lattice** $\Lambda \subseteq \mathbb{R}^d$ is any discrete additive subgroup of the Euclidean space $\mathbb{R}^d$;*

(ii) *the **rank** $rk(\Lambda)$ of Λ is the dimension of the linear space spanned by elements of Λ;*

(iii) *if $rk(L) = d$, we say L has a **full rank**;*

(iv) *if Λ' is a lattice and $\Lambda' \subseteq \Lambda \subseteq \mathbb{R}^d$, we say Λ' is the **sublattice** of Λ.*

Example. *Here are a couple of examples of lattices:*

(i) *$\mathbb{Z}^d$ is the full rank lattice of dimension d;*

(ii) *$\Lambda := \{\lambda_1(1,2) + \lambda_2(2,1) : \lambda_1, \lambda_2 \in \mathbb{Z}\} \subseteq \mathbb{R}^2$ is a full rank lattice;*

(iii) *In general, for $V = (v_1, \dots, v_k) \in \mathbb{R}^{d \times k}$ (matrix of full column rank), the lattice $\Lambda := \mathcal{L}(V) \triangleq \{\sum_{i=1}^k \lambda_i v_i : \forall_{i \in [k]}\ \lambda_i \in \mathbb{Z}\}$ is of rank k in $\mathbb{R}^d$;*

(iv) *$q\mathbb{Z}^d$ is a full rank sublattice of $\mathbb{Z}^d$; a lattice Λ such that $q\mathbb{Z}^d \subseteq \Lambda \subseteq \mathbb{Z}^d$ is called a **q-ary lattice**; these are extensively studied in lattice-based cryptography.*

We also make a couple of remarks without proof:

(a) if $T : \mathbb{R}^d \to \mathbb{R}^d$ is an invertible linear transformation and $\Lambda \subseteq \mathbb{R}^d$ is a lattice, then $\Lambda' := T(\Lambda)$ is also a lattice and moreover $\mathrm{rk}(\Lambda) = \mathrm{rk}(\Lambda')$;

(b) if Λ is a lattice, then $\mathbb{R}^d/\Lambda$ is a smooth manifold with a natural Lebesgue measure (call it μ) induced from $\mathbb{R}^d$;

(c) if Λ is also of full rank, $\mathbb{R}^d/\Lambda$ is compact and thus has a volume $\mu(\mathbb{R}^d/\Lambda)$ which we call the **covolume** of Λ (and denote $\mathrm{covol}(\Lambda)$).

Now we wish to classify the lattices. Turns out that any lattice Λ can always be generated by the $\mathbb{Z}$-span of finite collection of linearly independent vectors. Moreover, one of the vectors $w \in \Lambda$ can be chosen to be *irreducible*, which means $w/n \notin \Lambda$ for any $n \in \mathbb{Z}_{\geq 2}$.

Lemma 1.2 (Fundamental theorem of lattices)**.** *If $\Lambda \subseteq \mathbb{R}^d$ is of rank k, then there exist linearly independent vectors $v_1, \dots, v_k \in \mathbb{R}^d$ such that $\Lambda = \mathcal{L}(v_1, \dots, v_k)$.*

During construction we will see that v_1 can be made irreducible. One consequence of this lemma is that any lattice of rank k is isomorphic to $\mathbb{Z}^k$.

Proof. Without loss of generality, assume that $\mathrm{rk}(\Lambda) = d \geq 1$, else we could pass to a smaller vector space and continue the argument.

Additionally, observe that Λ contains at least one irreducible vector w: indeed, take any non-zero vector $v \in \Lambda$ and take w to be the smallest of the form v/n (such a vector must exist since Λ is discrete). By the full rank assumption, we find d linearly independent vectors $v_1, \dots, v_d \in \Lambda$ (with $v_1 := w$), so in particular the

volume of parallelepiped $|\bigwedge_{i=1}^{d} v_i| > 0$ (*notice that here* $\mathrm{span}(v_1,\dots,v_d) = \mathbb{R}^d$, *but* $\mathcal{L}(v_1,\dots,v_d) \neq \Lambda$). Since $\Lambda_0 := \mathcal{L}(v_1,\dots,v_d) \subseteq \Lambda$ is a sublattice with covolume $|\bigwedge_{i=1}^{d} v_i|$, we have $|\bigwedge_{i=1}^{d} v_i| \geq \mu(\mathbb{R}^d/\Lambda)$.

We proceed using the method of descent. We will build the sequence of sublattices $\{\Lambda_n\}_{n\in\mathbb{Z}}$ (with $\Lambda_n \subseteq \Lambda$) with the strictly decreasing covolumes (by a factor of $1/2$). Since we have a positive lower bound, we will conclude that $\Lambda_n = \Lambda$ at some point.

Here how it goes: consider the fundamental parallelipiped $\mathcal{P}(\Lambda_0) = \{\sum_{i=1}^{d} \lambda_i v_i : 0 \leq \lambda_i < 1\}$. Notice that it contains some non-zero point $x \in \Lambda$. Thus, write $x = \sum_{i=1}^{d} \lambda_i v_i$. Notice that at least one of $\lambda_2,\dots,\lambda_d$ is non-zero: otherwise, we would have $x = \lambda_1 w$ for $\lambda_1 \in [0,1)$, contradicting the fact that w is irreducible. Wlog, assume that $\lambda_d \neq 0$.

Now, we claim that we can assume $\lambda_d \leq 1/2$, so that our descent is fast enough. Indeed, assuming $\lambda_d > 1/2$, we can pick a different point from x: say, $x' = \sum_{i=1}^{d} v_i - x$, where the corresponding coefficient is less than $1/2$. Now set $\Lambda_1 := \mathcal{L}(v_1,\dots,v_{d-1},x)$. Notice that $\mathrm{covol}(\Lambda_1) \leq \frac{1}{2}\mathrm{covol}(\Lambda_0)$.

Define Λ_n from Λ_{n-1} recursively using the same steps. At some point we terminate and thus get the required $\mathbb{Z}$-basis.

Lemma 1.3 (Corollary). *This lemma has a couple of almost direct corollaries:*

(a) ***Splitting lemma.*** *Let Λ be a lattice of rank k, and let $v \in \Lambda$ be an irreducible vector. Then, there exists a sublattice $\Lambda' \subseteq \Lambda$ of rank $k-1$ such that $\Lambda = \Lambda' + \mathbb{Z}v$ with $\Lambda' \cap \mathbb{Z}v = \emptyset$.*

(b) ***Fundamental theorem of f.g. torsion-free additive groups.*** *Let Z be a finitely generated torsion-free additive group. Then, $Z \cong \mathbb{Z}^d$ for some $d \geq 0$.*

Proof.

Part (a). Apply Lemma 1.2 with $v_1 := v$, and set $\Lambda' := \mathcal{L}(v_2,\dots,v_k)$.

Part (b). Since Z is finitely generated, let $Z = (z_1,\dots,z_n)$ (for $z_i \in Z$). Define the lattice $\Lambda := \{u \in \mathbb{Z}^n : \sum_{i=1}^{n} u_i z_i = 0\}$. Now notice that it is precisely the kernel of the homomorphism $(u_1,\dots,u_n) \mapsto \sum_{i=1}^{n} u_i z_i$. Therefore, it induces the isomorphism $Z \cong \mathbb{Z}^n/\Lambda$. In particular, $\mathbb{Z}^n/\Lambda$ is torsion-free.

Now we proceed by induction. Base case is trivial and assume $\Lambda \neq \{0\}$. Pick some irreducible vector $v \in \Lambda$. Note that it is also irreducible in $\mathbb{Z}^n$: if $v = mw$ for some $w \in \mathbb{Z}^n$ and $m > 1$, then $w + \Lambda$ would be a non-zero element of $\mathbb{Z}^n/\Lambda$ such that $m(w+\Lambda) = 0 + \Lambda$, contradicting the torsion-free assumption for $\mathbb{Z}^n/\Lambda$. This implies that $\mathbb{Z}^n/v\mathbb{Z} \cong \mathbb{Z}^{n-1}$. We finish the proof by noticing that

$$\mathbb{Z}^n/\Lambda \cong \frac{\mathbb{Z}^n/(v\mathbb{Z})}{\Lambda/(v\mathbb{Z})} \cong \mathbb{Z}^{n-1}/\Lambda'.$$

2 Quotients of lattices

As already demonstrated in Lemma 1.3 (*part (b)*), quotiening two lattices is a powerful tool. More generally, if G is a finitely generated additive group by d elements $v_1,\dots,v_d \in G$, then $G \cong \mathbb{Z}^d/\Lambda_G$ where $\Lambda_G := \{u \in \mathbb{Z}^d : \langle u, v\rangle = 0\}$. Therefore, we provide the following tool to help analyzing quotients of two lattices.

Theorem 2.1 (Smith normal form). *Let $\Lambda, \Lambda' \subseteq \mathbb{R}^d$ be two full-rank lattices and $\Lambda' \subseteq \Lambda$. Then, there exist linearly independent $v_1, \ldots, v_d \in \Lambda$ such that*

$$\Lambda = \mathcal{L}(v_1, \ldots, v_d), \quad \Lambda' = \mathcal{L}(N_1 v_1, \ldots, N_d v_d),$$

with $N_1 \geq 1$ and $N_j \mid N_{j+1}$ for all $1 \leq j < d$.

Proof. We induce on d with $d = 0$ being vacuously true. For any non-zero $v \in \Lambda$ define the *index* of v (call it $\mathrm{ind}_\Lambda(v)$) as follows:

$$\mathrm{ind}_\Lambda(v) = \max\{n \in \mathbb{N} : v/n \in \Lambda\}.$$

Note that since Λ is discrete, $\mathrm{ind}_\Lambda(v) < \infty$. Also note that $\mathrm{ind}_\Lambda(v) = n$ iff $v = nw$ for irreducible $w \in \Lambda$.

Since Λ' has full rank, it contains non-zero vectors, each of which has an index. Let $N_1 := \min_{v \in \Lambda'} \mathrm{ind}_\Lambda(v)$. By the well-ordering principle there exists an irreducible vector $v_1 \in \Lambda$ such that $N_1 v_1 \in \Lambda'$.

Using Lemma 1.2, we may apply an invertible transformation to map Λ to $\mathbb{Z}^d$ such that v_1 gets mapped to the standard basis vector e_1. Now let $(n_1, \ldots, n_d)$ be any vector in Λ'. Observe that $n_1, \ldots, n_d$ are integers; furthermore, n_1 must be a multiple of N_1. Thus, we may factorize $\Lambda' = (N_1 e_1)\mathbb{Z} + \Lambda''$ where $\Lambda'' \subseteq \mathbb{Z}^{d-1}$ is a lattice. Also note that if $x \in \Lambda''$, then $(N_1, x) \in \Lambda'$ and hence (since (N_1, x) must have an index of at least N_1), x must be a multiple of N_1. Thus, $\Lambda'' \subseteq N_1 \mathbb{Z}^{d-1}$, so we may write $\Lambda' = N_1(e_1\mathbb{Z} + \Lambda''')$ for some sublattice $\Lambda''' \subseteq \mathbb{Z}^{d-1}$. Clearly $\mathrm{rk}(\Lambda''') = d - 1$.

We now invoke the inductive hypothesis. By applying an invertible transformation to $\mathbb{Z}^{d-1}$ if necessary, we may assume that

$$\Lambda''' = \{(n_2 M_2, \ldots, n_d M_d) : n_2, \ldots, n_d \in \mathbb{Z}\}$$

for some $1 \leq M_2 \leq \cdots \leq M_d$ such that $M_j \mid M_{j+1}$ for every j. The claim follows by setting $N_j := N_1 M_j$ for $j > 1$.

Lemma 2.2 (Corollary). *This lemma also has two nice corollaries:*

(a) ***Fundamental theorem of finite additive groups.*** *Every finite additive group G is isomorphic to the direct sum of a finite number of cyclic groups $\mathbb{Z}/N\mathbb{Z}$.*

(b) ***Fundamental theorem of f.g. additive groups.*** *Every finitely generated additive group G is isomorphic to the direct sum of a finite number of cyclic groups $\mathbb{Z}/N\mathbb{Z}$, and a lattice $\mathbb{Z}^d$ for some $d \geq 0$.*

Proof.

Part (a). Since the group is finitely generated, let $G = (g_1, \ldots, g_d)$. Define the map $\phi : \mathbb{Z}^d \to G$ as $\phi(n_1, \ldots, n_d) = \sum_{i=1}^d n_i g_i$. Using first isomorphism theorem, we as before note that $G \cong \mathbb{Z}^d / \ker(\phi) \leq \mathbb{R}^d / \ker(\phi)$. Clearly, $\ker(\phi)$ is a lattice of some rank $0 \leq k \leq d$. Moreover, it must be of full rank, otherwise $\mathbb{Z}^d / \ker(\phi)$ would be infinite, and thus so would hold for G. Hence by Lemma 1.2 it is generated by d linearly independent vectors $v_1, \ldots, v_d \in \mathbb{Z}^d$. Using the Smith normal form, we can after applying an isomorphism write $\ker(\phi)$ as the lattice generated by $N_1 e_1, \ldots, N_d e_d$

for some $N_1, \ldots, N_d \geq 1$; this makes G isomorphic to $G \cong \bigoplus_{i=1}^{d} (\mathbb{Z}/N_i\mathbb{Z})$ as desired (moreover, $N_i \mid N_{i+1}$ for each $i \geq 1$).

Part (b). Let $\mathrm{tor}(G) := \{x \in G : \exists_{n>0}\ nx = 0\}$ be the torsion group of G. By *part (a)* we know that $\mathrm{tor}(G)$ is the direct sum of cyclic groups. The quotient group $G/\mathrm{tor}(G)$ is torsion-free and is thus isomorphic to $\mathbb{Z}^d$ for some $d \geq 0$ by Corollary 1.3. Let $\widehat{e_1}, \ldots, \widehat{e_d}$ be arbitrary representatives in G of the standard basis $e_1, \ldots, e_d$ of $\mathbb{Z}^d$. Then, $G \cong \mathrm{tor}(G) \oplus \bigoplus_{i=1}^{d} \mathbb{Z}\widehat{e_i} \cong \mathrm{tor}(G) \oplus \mathbb{Z}^d$.

3 Convex bodies

Denote by $\mu(A)$ the volume of a set $A \subseteq \mathbb{R}^d$. If $A \in \mathbb{R}^d$ and $\lambda \in \mathbb{R}$, denote $\lambda \cdot A \triangleq \{\lambda a : a \in A\}$. Observe that $\mu(\lambda \cdot A) = |\lambda|^d \mu(A)$.

Definition 3.1. *We give a couple of new definitions:*

*(a) a set $A \subseteq \mathbb{R}^d$ is **convex** if $(1-\theta)x + \theta y \in A$ whenever $x, y \in A$ and $\theta \in [0,1]$;*

*(b) a set A is called a **convex body** if it is convex, open, non-empty, and bounded;*

*(c) a convex body A is **symmetric** if $A = -A$.*

Remark. *One can show that Definition 3.1 is equivalent to the following: the set A is convex iff $\lambda \cdot A + \mu \cdot B = (\lambda + \mu) \cdot A$ for all $\lambda, \mu \geq 0$. In particular, for convex sets we have $nA = |n| \cdot A$ for any $n \in \mathbb{Z}$.*

Notice that $\mu(A + A) = \mu(2 \cdot A) = 2^d \mu(A)$. For $A - A$, we can use

Lemma 3.2. *For any bounded open subsets $X, Y, Z \subseteq \mathbb{R}^d$ (not necessarily convex):*

$$\mu(X - Z)\mu(Y) \leq \mu(X - Y)\mu(Y - Z).$$

Using this lemma (set $X := A$, $Y := -A$, $Z := A$), we have $\mu(A - A) \leq 4^d \mu(A)$. In the converse direction, Brunn-Minkowski inequality gives $\mu(A - A) \geq 2^d \mu(A)$ (which we will hopefully discuss in the future).

Theorem 3.3 (John's Theorem). *Let $A \subseteq \mathbb{R}^d$ be a convex body. Then there exists an invertible linear map $T : \mathbb{R}^d \to \mathbb{R}^d$ and a point $x_0 \in A$ such that $B_d \subseteq T(A - x_0) \subseteq d \cdot B_d$, where $B_d = \{x \in \mathbb{R}^d : \|x\|_2 < 1\}$. If A is symmetric, then we can improve these inclusions to $B_d \subseteq T(A) \subseteq \sqrt{d} \cdot B_d$.*

Proof sketch. Define the *ellipsoid* to be any set E of the form $E = L(B_d) + x_0$ where $x_0 \in \mathbb{R}^d$ and L any linear map (not necessarily invertible).

Since A is open and bounded, it is easy to see that the set of all ellipsoids E contained in A is a compact set. Besides, the volume of ellipsoid E is $\mu(E) = |\det L|$, which is clearly a continuous function of E. Thus there exists an ellipsoid $E = L(B_d) + x_0$ in A that maximizes the volume $\mu(E)$; since A is open, this volume is non-zero, and hence L is invertible. By applying L^{-1}, we may thus assume that E is a translate $E = B_d + y_0$ of a unit ball, where $y_0 = L^{-1}(x_0)$. Wlog, now assume that we have such an E in A, keeping in mind that $T = L^{-1}$.

Assume for now that A is symmetric. Observe that since A contains $B_d + y_0$, then

it also contains $B_d - y_0$ by symmetry, and hence contains B_d since it is in the convex hull of $B_d + y_0$ and $B_d - y_0$. To conclude the proof of the lemma in this case we need to show that $A \subseteq \sqrt{d} \cdot B_d$.

Suppose for contradiction this is not the case, so suppose wlog that $re_1 \in A$ for some $r > \sqrt{d}$ and e_1 the first basis vector. The rest of the proof consists in constructing some ellipsoid L contained in A but with $\mu(L) > \mu(B_d)$ (contradicting the minimality of B_d). Specifically,

$$L := L_{\varepsilon,\delta}(B_d), \quad L_{\varepsilon,\delta} := \begin{pmatrix} 1 + (\sqrt{d} - 1 + \delta)\varepsilon & 0 \\ 0 & (1-\varepsilon)\mathbb{I}_{d-1} \end{pmatrix}.$$

A couple of statements that finishes the proof:

1. $L_{0,\delta}(B_d) = B_d$ (that is, for $\varepsilon = 0$);
2. $\det L_{\varepsilon,\delta} = (1-\varepsilon)^{d-1}(1 + (\sqrt{d} - 1 + \delta)\varepsilon)$ has a positive ε-derivative at $\varepsilon = 0$;
3. for small enough δ, we have $L_{\varepsilon,\delta}(B_d) \subseteq A$.

For symmetric case turns out that we can use the same transformation $L_{\varepsilon,\delta}$, but to ensure that the third condition is met, we need the radius of d for the ball.

References

[TV06] Terence Tao and Van H. Vu. *Additive Combinatorics*. Cambridge Studies in Advanced Mathematics. Cambridge University Press, 2006.